

		Since both r and ω are constants, centripetal acceleration does not change over time.
	10	A Conservation of energy gives $2400 \times 120 = 0.106 \times 2260000 + \text{heat loss}$. Heat loss = 48 440 J. So rate of heat loss = $\frac{48440}{120} = 404 = 400 \text{ W}$
	11	C density of the gas 6.80×10^{15} 2.02×10^{-3} $=$ $1.37 \times 10^{-2} \text{ kg m}^{-3}$